

CMP-10 - Learning and Wisdom System

Convert governed experience into reproducible patterns and bounded wisdom while preventing silent self-modification of protected behavior.

Authority and state ownership

Book I: Decisions 8-23, 30-37, 40-43, 47-50, and 58. Book II: Chapters 11, 12. Authoritative state: Experience Ledger, Pattern Records, Wisdom Records, learning-run records, promotion decisions. Explicitly non-owned: constitutional meaning, authority, deployment into a declared live-effect environment and approved Reality Boundary stage, Knowledge Graph.

Critical invariant

Learning cannot silently modify constitutional meaning, authority, protected behavior, or authoritative live-effect state.

Logical interfaces

CMP-10-IF-01 / Command / Record Experience

Command `Record Experience` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-10-IF-02 / Command / Propose Pattern

Command `Propose Pattern` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-10-IF-03 / Command / Promote Pattern

Command `Promote Pattern` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-10-IF-04 / Command / Retire Wisdom

Command `Retire Wisdom` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-10-IF-05 / Query / Get Experience

Query `Get Experience` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-10-IF-06 / Query / Explain Pattern

Query `Explain Pattern` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-10-IF-07 / Query / Get Wisdom Limits

Query `Get Wisdom Limits` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-10-IF-08 / Event / Experience Recorded

Event `Experience Recorded` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-10-IF-09 / Event / Pattern Promoted

Event `Pattern Promoted` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-10-IF-10 / Event / Wisdom Retired

Event `Wisdom Retired` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

Lifecycle, controls, and recovery

Lifecycle: Experience: Recorded > Validated > Retained; Pattern: Candidate > Qualified > Approved > Retired. All transitions require version, guard, identity, authority where applicable, idempotency, time context, and evidence. Contain: training-data poisoning, pattern overgeneralization, silent promotion, feedback loop, unbounded self-modification. Recovery proves identity, state, authority, provenance, dependencies, and invariants before mutation resumes.

Conformance tests

CMP-10-TST-001 - Sole-owner mutation. Attempt mutation through a consumer, replica, cache, operator path, and direct datastore route. Expected: Every unauthorized path is denied and evidenced.

CMP-10-TST-002 - Authority denial. Invoke every command with absent, expired, revoked, mismatched, and over-broad authority context. Expected: No state changes; denial reason and evidence are complete.

CMP-10-TST-003 - Valid lifecycle. Exercise every declared transition with valid guards and fresh versions. Expected: One transition Evidence Object is admitted per transition identifier; duplicate attempts produce no additional state mutation or admitted Evidence Object.

CMP-10-TST-004 - Invalid lifecycle. Attempt skipped, stale, replayed, and terminal-state transitions. Expected: Every invalid transition is rejected without partial mutation.

CMP-10-TST-005 - Critical invariant. Learning cannot silently modify constitutional meaning, authority, protected behavior, or authoritative live-effect state. Expected: Invariant holds under normal, concurrent, degraded, and recovery conditions.

CMP-10-TST-006 - Failure containment. Inject training-data poisoning, pattern overgeneralization, silent promotion, feedback loop, unbounded self-modification. Expected: Unsafe work stops, evidence is preserved, and declared containment state is observable.

CMP-10-TST-007 - Recovery. Restore from a verified checkpoint, reconcile dependencies, and resume after validation. Expected: Identity, state, authority, provenance, and evidence continuity are proven.

CMP-10-TST-008 - Contract compatibility. Run current and prior supported contract versions plus an unsupported version. Expected: Supported behavior remains compatible; unsupported input fails explicitly.

CMP-10-TST-009 - Privacy and security. Inject secrets, personal data, malicious input, confused-deputy requests, and log-exfiltration probes. Expected: Least privilege and data controls hold; prohibited data is absent from telemetry.

CMP-10-TST-010 - Topology independence. Repeat critical tests under permitted single-node, replicated, partitioned, and restored topologies. Expected: Topology does not multiply authority or mutation ownership.

Traceability and certification

Twelve requirements map to Books I-III and Book X. No open finding or Owner Review item. Final for Book IV v1.0.